

1. Administrative & Study Identification

Full Study Title: A 2 Year, Double-blind, Randomized, Multicenter, Active-controlled Core Phase to Evaluate Safety & Efficacy of Daily Fingolimod vs Weekly Interferon β -1a im in Pediatric Patients With Multiple Sclerosis and 5 Year Fingolimod Extension Phase **Short Title/Acronym:** PARADIGMS Study (Safety and Efficacy of Fingolimod in Pediatric Patients With Multiple Sclerosis) **Protocol Number:** CFTY720D2311 **NCT Number:** NCT01892722 **Posting ID:** 51998400994137915 **Sponsor/Company Name:** Novartis **Investigational Product:** Fingolimod (Gilenya) vs Interferon beta-1a **Phase of Development:** PHASE3 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Novartis Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of oral fingolimod compared to intramuscular interferon beta-1a in reducing the annualized relapse rate (ARR) in children and adolescents with a confirmed diagnosis of relapsing multiple sclerosis (MS) over a period of up to 24 months. **Secondary Objectives:** To evaluate the annualized rate of new or newly enlarged lesions on T2-weighted magnetic resonance imaging (MRI), the time to first relapse, the proportion of patients remaining relapse-free, and the time to confirmed disability progression.

2.2 Background & Rationale To address the unmet medical need for highly effective, approved, and well-tolerated oral disease-modifying therapies (DMTs) in pediatric MS. Pediatric MS is associated with frequent and severe relapses that lead to early physical and cognitive disability accumulation compared to adult-onset MS. The study includes a Core Phase (24-month, double-blind, randomized) and an Extension Phase (60-month, 5-year phase). A younger cohort (up to 25 patients) fulfilling specific age (≤ 12 years), weight (≤ 40 kg), or pubertal criteria (Tanner stage < 2) was also requested as a post-approval health authority commitment to participate in the Extension Phase.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (interferon beta-1a) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$N\$**: 240 children and adolescents **Number of Sites**: 87 sites across 25 countries **Estimated Study Start**: July 2013 **Estimated Primary Completion**: August 2017 (for the 24-month core phase), with an open-label extension phase continuing to an estimated completion in July 2029.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria Core Phase: 1. Age 10 to 17 years at the time of screening. 2. Confirmed diagnosis of multiple sclerosis (MS). 3. At least one MS relapse during the previous year or two MS relapses in the previous 2 years, or evidence of Gadolinium (Gd) enhancing lesions on MRI within 6 months. 4. Expanded Disability Status Scale (EDSS) score of 0 to 5.5, inclusive.

Extension Phase: 1. Patients that originally met Core Phase Inclusion criteria and completed the Core phase on or off of study drug. 2. Newly recruited patients enrolling directly into the Extension Phase (including the younger cohort of up to 25 patients) must fulfill the local country health authority product label approved for the pediatric age group. 3. Central review (including initial MRI report) of the diagnosis of pediatric MS is required for all newly recruited patients.

4.2 Exclusion Criteria Core Phase: 1. Patients with progressive MS. 2. Patients with an active, chronic disease of the immune system other than MS. 3. Patients meeting the definition of Acute Disseminated Encephalomyelitis (ADEM). 4. Patients with severe cardiac disease or significant findings on the screening ECG. 5. Patients with severe renal insufficiency.

Extension Phase: 1. Premature discontinuation of the study drug during the Core Phase due to an adverse event (AE), serious adverse event (SAE), laboratory abnormality, or other conditions leading to permanent study drug discontinuation due to safety reasons. 2. Patients with known new events or concomitant medications (washout periods required prior to Visit 15) that would exclude them from the Core Phase exclusion criteria. (Serological or other additional tests will not be required.) 3. Newly recruited patients in the younger cohort must fulfill the exclusion criteria for the core phase.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral fingolimod administered once daily (0.5 mg for body weight >40 kg; 0.25 mg for body weight ≤ 40 kg). The active control arm received intramuscular interferon beta-1a at 30 µg once weekly. **Route of Administration**: Oral (fingolimod) vs. Intramuscular injection (interferon beta-1a). **Duration of Treatment**: 24 months for the

double-blind core phase. Patients completing the core phase (and newly recruited patients in the younger cohort) were eligible to enter a 60-month (5 year) open-label extension phase receiving fingolimod.

5.2 Concomitant & Prohibited Therapy Permitted: Short courses of corticosteroids for the acute management of objective MS relapses. **Prohibited:** Concurrent use of other immunosuppressive medications or alternative disease-modifying therapies (DMTs) during the core double-blind phase.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	2-4 weeks prior	Informed Consent/ Assent, Medical History, Neurological Exam, ECG, Baseline MRI.
Baseline	Day 0	Randomization, First Dose Administration (with continuous cardiac monitoring due to first-dose bradycardia risk).
Interim	Q3-6 Months	Clinical relapse assessment, EDSS scoring, Safety Labs (e.g., leukocyte counts), and Follow-up MRIs (at 6, 12, and 24 months).
End of Study	Month 24	Final efficacy assessment, Transition to Extension Phase or Study Exit/Safety follow-up.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Frequency of relapses assessed by the annualized relapse rate (ARR) in patients treated for up to 24 months. **Measurement Method:** The ARR is defined as the average number of confirmed relapses per year (total number of confirmed relapses divided by the total days in the study multiplied by 365.25), evaluated by an independent rater.

7.2 Secondary & Exploratory Endpoints * Annualized rate of new or newly enlarged lesions on T2-weighted magnetic resonance imaging (MRI). * Time to first relapse. * Proportion of patients remaining relapse-free. * Time to 3-month and 6-month confirmed disability progression (EDSS). * Safety and tolerability profile, including adverse events (AEs) and serious adverse events (SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring, ECGs, ophthalmologic exams, and laboratory tracking. Specific focus on leukopenia, headache, flu-like symptoms, and cardiovascular events. **Serious Adverse Events (SAEs):** Reported promptly to the sponsor. Discontinuation due to AEs, SAEs, or lab abnormalities permanently excluded patients from entering the Extension Phase. **Data Monitoring Committee (DMC):** An independent

Data Monitoring Committee oversaw the safety and trial integrity throughout the duration of the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses and Safety Set for the evaluation of AEs/SAEs. **Sample Size Justification:** Powered to detect a statistically significant superiority (reduction in ARR) of fingolimod compared to interferon beta-1a. **Handling of Missing Data:** Adjusted analyses utilized negative binomial regression models for ARR calculation, accommodating varied time-in-study due to early dropouts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits by the sponsor to ensure protocol adherence, accurate data entry, and safety reporting. **Audit Trail:** Robust clinical data management systems were utilized with strict eCRF locking procedures and data clarification protocols. Central review (including initial MRI report) of the diagnosis of pediatric MS was required for all newly recruited patients.

11. Study Identifiers & Governance

Primary ID: CFTY720D2311 **Registry Number:** NCT01892722 **Posting ID:** 51998400994137915 **Sponsor/Lead Organization:** Novartis **Study Title:** Safety and Efficacy of Fingolimod in Pediatric Patients With Multiple Sclerosis **Official Regulatory Title:** A 2 Year, Double-blind, Randomized, Multicenter, Active-controlled Core Phase to Evaluate Safety & Efficacy of Daily Fingolimod vs Weekly Interferon β -1a im in Pediatric Patients With Multiple Sclerosis and 5 Year Fingolimod Extension Phase

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Pediatric Relapsing Multiple Sclerosis (MS) **Disease Area:** Multiple Sclerosis **Diagnosis Threshold:** Confirmed MS diagnosis with at least 1 recent relapse (or 2 in 2 years/Gd-enhancing lesions within 6 months) and an EDSS score of 0-5.5 **Medical Methodology:** Disease-Modifying Therapy (DMT) application **Optimization Target:** Reduction of Annualized Relapse Rate (ARR) and suppression of central nervous system inflammatory activity (MRI lesions)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Pediatric MS monitoring and relapse intervention. **Education Protocol (HCP):** Specialized site training regarding pediatric MS scoring (EDSS) and S1P receptor modulator first-dose cardiac observation. **Education Protocol (Patient):** Training on daily oral medication adherence, potential side effects, and recognition of relapse symptoms. **Quality Audit Mechanism:** Blinded independent radiological central review for MRI outcomes and independent evaluating neurologists for relapse confirmation.

14. Observation & Follow-Up Schedule

Total Duration: 24 months (Core Phase) followed by a 60-month (5 Year) open-label Extension Phase **Onsite Visit Cadence:** Month 1, Month 3, and every 3-6 months thereafter for routine clinical exams and MRI **Remote Monitoring Frequency:** Regular telephone check-ins between clinic visits for safety and AE screening **Checklist Review Interval:** Every visit for compliance tracking and medication accountability

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				